The term neuropsychology in its English version originated quite recently, in part because it represented a new approach to studying the brain. The mastermind of all our living being is the brain which controls every aspect of behaviour from molar to molecular, a universe within the small cranium.

Millions of neurons and glial cells interaction, firing constantly resulting in the gross behaviours we see or experience. We put together a definition encompassing these and simply state that "Behavioural Neuro-sciences aims to study brain-behaviour relationships utilizing all its specializations and methodologies".

Brief historical background

Behavioural Neuro-sciences is as wide as the fields within the scope of its domain and every single aspect of brain/behaviour relationship of any organisms comes under its purview.

- **Rene Descartes'** speculations about the roles of the mind and brain in the control of behaviour provide a good starting point in the history off physiological psychology. To Descartes, animals were mechanical devices; their behaviour was controlled by environmental stimuli. His view of the human body was much the same: it was a machine. Reactions like this did not require participation of the mind; they occurred Automatically. He called them **reflexes**.
- Descartes was a **duellist**, he believed that each person possesses a mind – a unique human attribute that is not subject to the laws of the universe. He was the first to suggest that a link exists between the human mind and its purely physical housing, the brain.
- He noted that the brain contains hollow chambers (the ventricles) that are filled with fluid, and he hypothesized that this fluid is under pressure. In his theory,when the mind decides to perform an action, it tilts the pineal body in a particular direction like a little joystick, causing fluid to flow from the brain into the appropriate set of nerves. This flow of fluid causes the same muscles to inflate and move.
- In science, a model is a relatively simple system that works on known principles and is able to do at least some of the things a more complex system can do.
- **Gialvin**: Galvin found that electrical stimulation of a frog's nerve caused contraction of the muscle to which it was attached. Contraction occurred even when the nerve and muscle were detached from the rest of the body, so the ability of the muscle to contract and the ability of the nerve to send a message to the muscle were characteristics of these tissues themselves. Thus, the brain did not inflate muscles by directing pressurized fluid though the nerve.
- **Johannes Muller** was a forceful advocate of the application of experimental techniques of physiology. In his doctrine of specific nerve energies, Muller observed that although all

nerves carry the same basic message an electrical impulse – we perceive the messages of different nerves in different ways

- **Flourens**: Flourens removed various parts of animals' brains and observed their behaviour. This method is called experimental abolition.
- **Paul Broca** applied the principle of experimental ablation to the human brain. He observed the behaviour of people whose brains had been damaged by strokes. In 1861 he performed an autopsy on the brain of a man who had had a stroke that resulted in the loss of the ability to speak.
- In **1870**, German physiologists Gustav Fritsch and Eduard Hitzig used electrical stimulation as a tool for understanding the physiology of the brain. They applied weak electrical current to the exposed surface of a dog's brain and observed the effects of the stimulation. They found that stimulation of different portions of a specific region of the brain caused contraction of specific muscles on the opposite side of the body.
- Hermann von Helmholtz devised a mathematical formulation of the law of conservation of energy, invented the ophthalmoscope (used to examine the retina of the eye), and devised an important and influential theory of colour vision and. Colour blindness, and studied audition, music, and many physiological processes.

Current trends in the Field of Behavioural Neuroscience

1. **Brain-machine interfaces**: Brain-machine interfaces (BMIs) ard devices that connect the brain to external machines, allowing people to control them using their thoughts. This technology has the potential to revolutionize fields such as prosthetics, robotics, and rehabilitation.
2. **Optogenetics**: Optogenetics is a technique that uses light to control the activity of neurons in the brain. This technology has the potential to reveal the neural basis of complex behaviours and to develop new treatments for neurological and psychiatric disorders.
3. **Social neuroscience**: Social neuroscience investigates the neural mechanisms underlying social behaviour and cognition, including empathy, theory of mind, and social decisionmaking. This field has the potential to shed light on the neural basis of social disorders such as autism spectrum disorder and social anxiety disorder.
4. **Brain plasticity**: Brain plasticity refers to the brain's ability to change and adapt in response to experience. Research in this field is focused on understanding the neural mechanisms of brain plasticity and developing interventions to enhance brain plasticity for the treatment of neurological and psychiatric disorders.

5. **Neuroimaging**: Neuroimaging techniques such as magnetic resonance imaging (MRI) and positron emission tomography (PET) allow researchers to visualize the structure and function of the brain in living humans. These techniques are increasingly being used to study the neural basis of behaviour and to develop new diagnostic and therapeutic approaches.

Contribution of various areas and related discipline

1. **Neuroscience**: Neuroscience is the study of the nervous system, including the brain, spinal cord, and peripheral nervous system. Neuroscience contributes to behavioural neuroscience by providing a foundation for understanding the biological basis of behaviour, cognition, and emotion. It also provides tools for investigating the structure, function, and development of the nervous system.

2. **Psychology**: Psychology is the scientific study of behaviour and mental processes. It contributes to behavioural neuroscience by investigating the relationship between behaviour and the brain. It also provides theories and models for understanding how cognitive and emotional processes work, and how they are related to brain function.

3. **Biology**: Biology is the study of living organisms and their interactions with each. Other and their environment. It contributes to behavioural neuroscience by providing knowledge of the cellular and molecular mechanisms that underlie brain function, development, and plasticity.

4. **Computer science**: Computer science is the study of computational systems, including software, hardware, and algorithms. It contributes to behavioural neuroscience by providing tools and techniques for analysing complex datasets, modelling brain function, and developing brain-machine interfaces.

6. **Mathematics**: Mathematics provides the language and tools for modelling and analysing complex systems, including the brain. It contributes to behavioural neuroscience by providing quantitative methods for analysing neural data, modelling neural circuits and behaviour, and developing computational models of brain function.

7. **Physics**: Physics is the study of the fundamental principles that govern the behaviour of matter and energy. It contributes to behavioural neuroscience by providing tools and techniques for imaging brain function, such as magnetic Darwinie resonance imaging (MRI), computed tomography (CT), and positron emission tomography (PET)

Major Contributors

Hippocrates stated that the brain was the seat and centre of sensation, thought, emotions and judgement.

Muslim contribution is the first recorded brain dissection with anatomical details were given by Muslim scientists. This is experimental as well as descriptive (not speculative work). While dissecting, they discovered the hard protective covering. Protecting the brain and named it as Umm ul Dafah (hard protective covering) and the inner covering also the fragile mother. These were translated verbatim into Latin during the renaissance as Dura Mater and Pia Mater (Mater from Umm mother, protective covering).

Franz Gall presented the concept of phrenology where faculties were located in centres of the brain. The bumps on the cranium were also part of his formal theory. However, he also presented the concept that the two hemispheres were joined by corpus collosum.

Around 1800's, **Flourens** was the first one to experiment with ablation of avian brains. He demonstrated loss of function with damage. He proposed the concept of equipotentiality of the brain on the basis of his investigations.

In 1861, **Paul Broca** presented evidence for speech expression in specific brain areas, i.e. frontal motor areas for speech. This area is formally known as the Broca's area now.

In1868, **Hughlins Jackson** presented the idea of differentiation of two types of language functions- expressive and receptive, he also elaborated on a particular form of epilepsy known as the Jacksonian seizure)

In late 1800's **Wernicke's** presented evidence for control of receptive speech in Synaptic Temporal lobe.

. Many eminent names followed up on. Brain research some of those who got Nobel Plasti prizes are Gazzaniga, Sperry for their work in the 1960's. Lashley and Hebb are known as the fathers of Behavioural Neuro-sciences as we know it today.

Darwinian Theory of Evolution

Evolutionary psychology is a scientific discipline that approaches human behaviour through a lens that incorporates the effects of evolution. It combines the science of psychology with the study of biology.

Evolutionary psychologists seek to explain people's emotions, thoughts, and responses based on Charles Darwin's Theory of Evolution Through Natural Selection, similarly to how evolutionary biologists explain an organism's physical features. Proponents of this psychological approach posit that as our ancestors confronted problems and developed ways of solving them, some had certain innate instincts and intelligence that gave them the ability to figure out and apply the most successful solutions.

Evolutionary psychology is a well-defined discipline of study and research, with fundamental foundations that have developed and continue to guide new studies. There are five basic principles of evolutionary psychology:

- Your brain Is a physical system that instructs you to behave in a manner appropriate and adaptive to your environment.
- The neural circuitry of your brain helps you solve problems in an appropriate manner. The specific ways that the neural circuitry is constructed were directed by natural selection, over the course of generations.
- Most of your psychological behaviours are determined subconsciously by your neural circuitry, and you are largely unaware of these subconscious processes.
- You rely on conscious decision-making to guide you in your daily life, and you may be aware of the conclusions resulting from the complex neural circuitry while remaining unaware of the underlying process involved.
- Neural circuits in the brain are specialized to solve different adaptive problems. For example, the circuitry involved in vision is not the same as for vomiting.